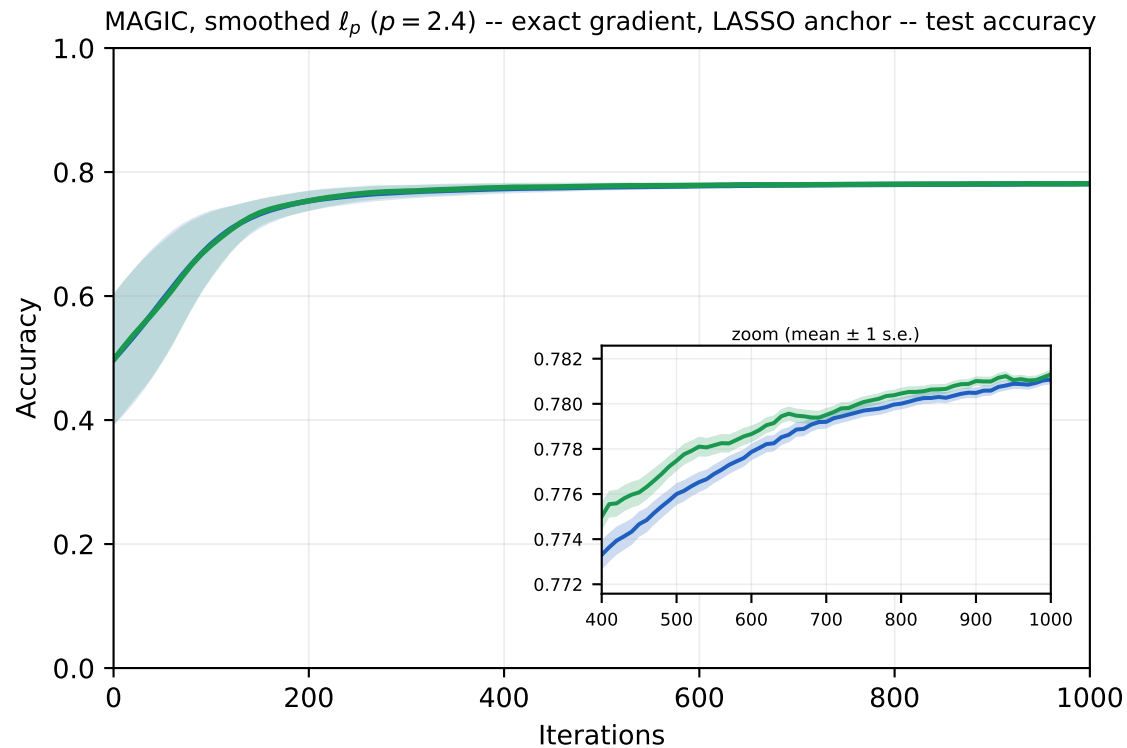
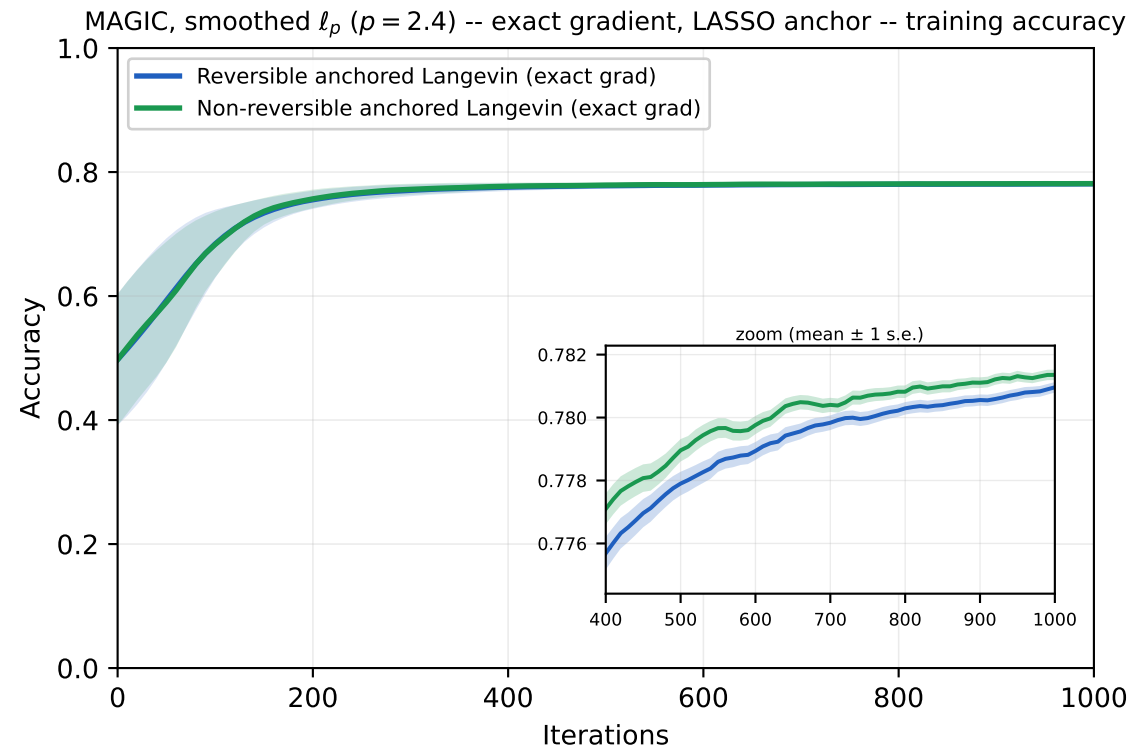


magic_lp: exact-gradient anchored Langevin, s = (5, 5, 5)



MAGIC -- smoothed ℓ_p $g(w) = \sum_i (w_i^2 + \text{eps}^2)^{p/2} \leq \text{Lambda}$, $p = 2.4$, $\text{eps} = 0.2$, $\text{Lambda} = 4$. $d = 10$ (intercept w_0 + 9 features). $U = \sum \text{softplus} - y \cdot x \cdot w + w_0^2 / (2 \sigma^2) + \text{lambda} \sum_{j \geq 1} |w_j|$; U_0 replaces $|w_j|$ by $\sqrt{w_j^2 + \text{delta}^2}$; $a = \exp(U - U_0)$. EXACT gradient (no mini-batching). $\sigma = 10$, $\text{lambda}_{\text{lasso}} = 152 = 0.01$. n_{train} , $\text{delta} = 0.000506$ (so a in $[0.50, 1]$). Block strengths $s = (5, 5, 5)$, acting on $w_1..w_9$ only. Step $\text{eta} = 1e-06$ (actually used). $R = 150$ independent replicates on one fixed stratified 80/20 split ($n_{\text{train}} = 15216$, $n_{\text{test}} = 3804$). Bands: mean $\pm$ 1 SD (ddof = 1) across replicates -- repeat-run variability at a fixed split, not a confidence or credible interval. eta and s selected by a TRAINING-ONLY sweep over 5 step sizes $\times$ 4 strengths; these curves are a CONFIRMATION run on independent replicates (seed 4100) not used for that selection. Verdict on the paired training difference: non-reversible WINS.